



◀ GLIDING AND CRAWLING

Many of the animals that attach themselves to the seafloor are eaten by other creatures that move over the bottom. Some, like these sea slugs with their violet-tipped tentacles, glide over the rocks to prey on attached animals. Sea snails, starfish, and sea urchins move in a similar way. Mobile crustaceans such as crabs and lobsters crawl about on well-developed legs, in search of prey and edible scraps. Most are weighed down by their heavy shells, but smaller shrimps and prawns can swim.

UNDERWATER SENSES



VISION

Light does not penetrate far under water, especially if the water is cloudy, but many animals have good vision for both hunting and defense. An octopus has eyes like ours, and the scallops that it eats have simple eyes dotted around their shell rims to detect danger.



SMELL

Scent and taste are vital clues to the presence of food. Sharks are famous for the way they detect tiny amounts of blood in the water, and many carnivorous sea snails, like this one, locate their victims by sniffing them out with their long, sensitive snouts.



HEARING

Sound travels fast under water, and for very long distances. Many animals such as fish can detect such sounds even though they do not have ears. Dolphins exploit this by using bursts of sound to scare fish into dense groups, making them easier to catch.



PRESSURE SENSE

Many oceanic animals are able to detect pressure changes in the water. Fish pick them up through a line of sensors along each flank, called the lateral line. Pressure changes make them aware of movements nearby, and enable shoals to swim in perfect unison.



ELECTRO-SENSITIVITY

Some marine creatures such as sharks and rays are able to detect the faint electrical nerve signals of other animals. The electro-sensors that pit the snout of this shark enable it to locate even invisible, scentless prey with deadly accuracy.

▲ SWIMMING

The bodies of many marine animals have almost the same density as water. This gives them neutral buoyancy, so they can hang in the water above the seafloor. Most fish have inflatable floats that they use to adjust their buoyancy. Others, such as sharks, sink slowly unless they keep swimming. Open-water fish like these mackerel are highly streamlined, so they can move fast without wasting energy.



◀ DEEP DIVING

Some air-breathing sea animals are specially equipped for diving. They include mammals, like this sperm whale, that dive to extreme depths to seize fish and squid. The main problem for these marine mammals is surviving the intense pressure at great depth.

This makes their lungs useless, so they have to store vital oxygen in their blood and muscles.